GIS Programming and Customization
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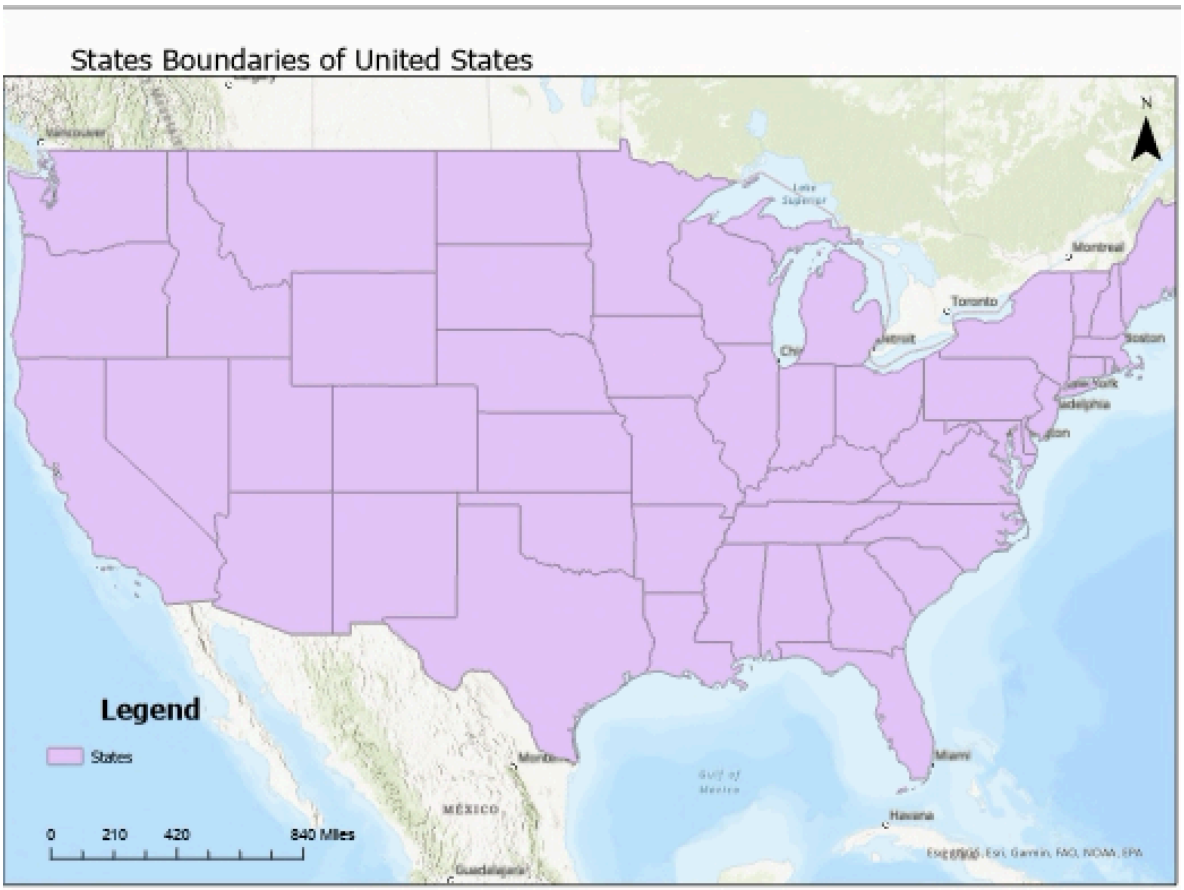
Gasoline Price Toolbox Project Report

Introduction.

The goal of this project is to build a geospatial data processing tool using Python to perform critical functions such as managing data collection, data cleaning and reformatting, data storage, analysis, modeling, and visualization via maps, graphs, tables. This project is created and developed based on project 4 that creates a Python toolbox in ArcGIS Pro that calculates the quantile of gasoline price. The script tool classifies numerical data into different categories by manipulation of the Excel data, then joining the attribute table of a geodatabase file with the Excel datasets, and creating a new feature layer based on the joined table that visualizes the distribution of retail price of gasoline.

Study area.

The study area is the United States of America. It is a country primarily located in North America. It consists of 50 states, a federal district, five major unincorporated territories,



326 Indian reservations, and nine minor outlying islands. The United States shares land borders with Canada to the north and Mexico to the south as well as maritime borders with the Bahamas, Cuba, and Russia, among others.

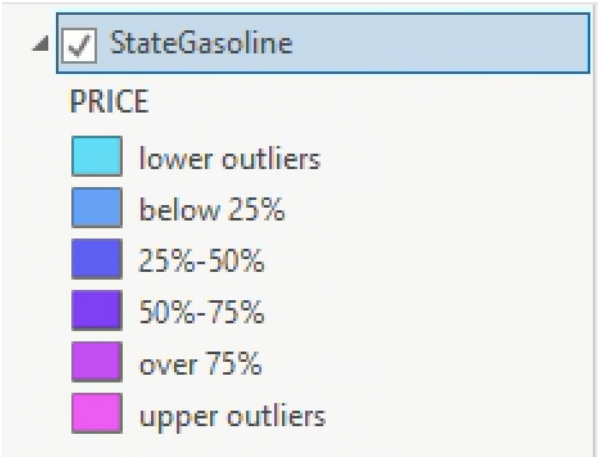
Data and Data Processing.

The dataset includes an excel file and a geodatabase file. The excel file contains the retail price of fuel gasoline in each state within the United States. The geodatabase file is the states boundaries. The spatial reference is NAD 1983. And the Excel data has one table sheet and includes two variables: County and Price. Since some states are not included in the gasoline data file, the data cleaning step needs to exclude the states : Puerto Rico, District of Columbia, American Samoa, and Guam.

Methods.

The first step is to create a new python toolbox. The toolbox is created by clicking the “Toolboxes” then “New Python Toolbox” in the Catalog pane of ArcGIS Pro. In the Toolbox class' __init__ method, properties of the toolbox are defined, including the alias, label, and description. The name of the toolbox is defined by the name of the .pyt file.

Min	3.63
Q1	3.78
Q2	3.95
Q3	4.15
Max	5.34



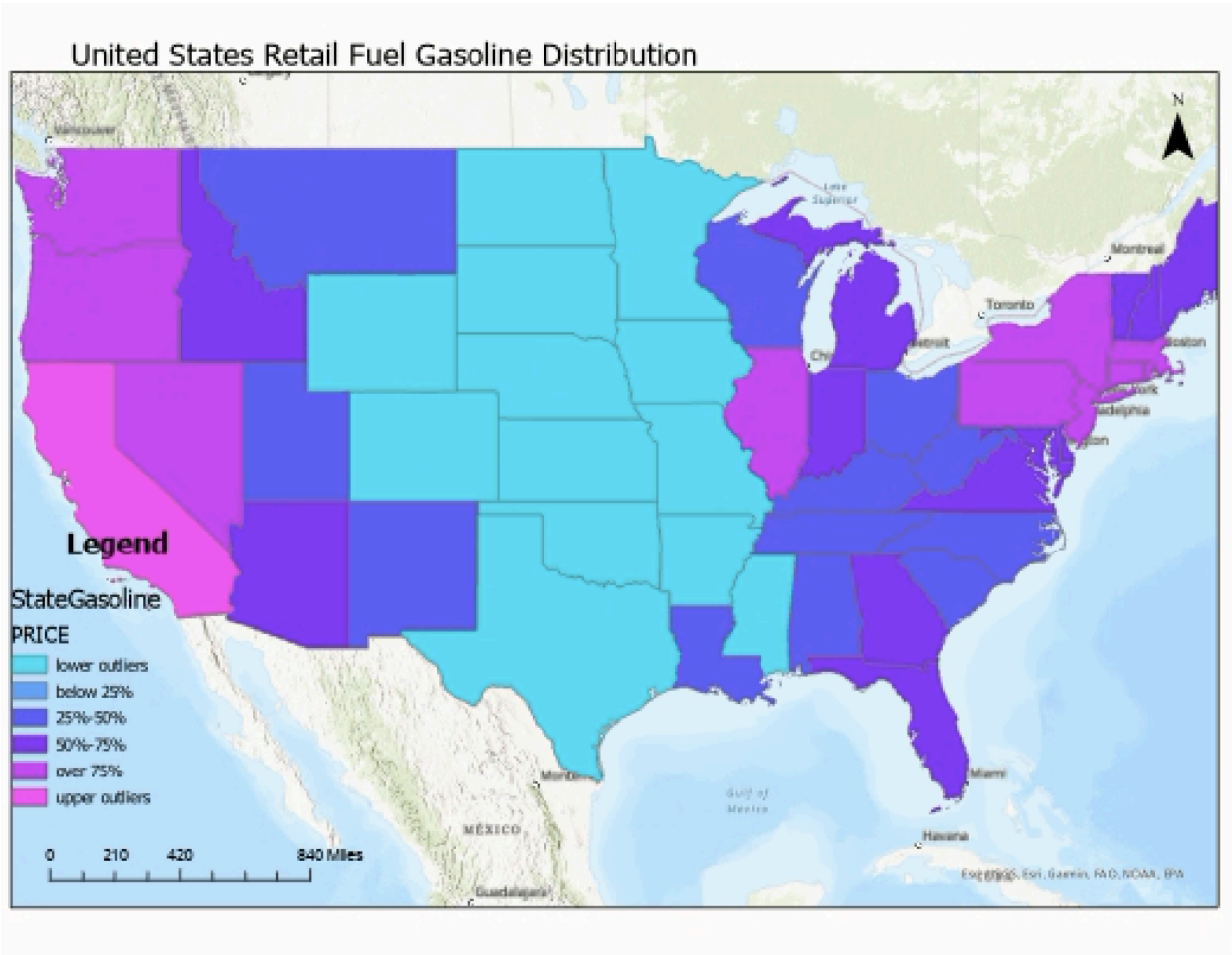
The second step is to manipulate the excel data. Use the DataFrame.quantile to calculate the Q1, Q2, Q3 and IQR of the gasoline retail price. Based on the quantiles results, classify numerical data into six categories. Then, create a new excel file by DataFrame.to_excel.

The third step is to join the attribute table of a geodatabase file and the excel file. The python code is arcpy.AddJoin_management(inFeatures, joinField, joinTable, joinField) that inFeatures is the geodatabase file, joinTable is the new Excel file. This line of code will call the Add Join tool in ArcGIS Pro to execute the join step. Based on the joined table, create a new feature layer by arcpy.CopyFeature_management function.

The last step is to set the symbology of the new layer to the manual interval and change the range of class according to the quantiles of data.

Results.

The results of toolbox generates the statistics quantile to classify the numerical data from Excel file into six different categories: outliers over Q3, less than 25%(Q1), greater than 75%(Q3), outliers below Q1, 25%-50%(Q1-Q2), and 50%-75% (Q2-Q3). Then use AddJoin and Copyfeature functions to join the Excel table and the attribute table of a geodatabase file and create a new feature layer based on the joined table.



For the toolbox created, the input feature is the geodatabase file Country Boundaries, the input table is the excel file retail gasoline price by states, the input field NAME is the name of each county.

After running the toolbox that joins the geodatabase file of Country Boundaries and the excel file of United States retail gasoline price by the field NAME. The new feature layer with classification visualized in the map below.

Discussion.

Overall, this project evaluates and visualizes the statistical distribution of retail gasoline price per unit by quantiles method. Also, this project demonstrated how the toolbox from project 4 can be modified. For the limitation of this project, the retail gasoline price is missing some state data as I mentioned in the data section. Therefore the distribution is not a hundred percent accurate.

References.

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